Apalutamide Tablets in Different Food Vehicles

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PURPOSE

- Apalutamide (ERLEADA®) is an androgen receptor inhibitor[1] currently approved in the United States and European Union for the treatment of non-metastatic castration-resistant prostate cancer and metastatic castration-sensitive prostate cancer.[2]
- The approved dose of apalutamide is 240 mg (4 x 60 mg tablets), to be taken once-daily, swallowed whole, with or without food.[2]
- Patients with difficulty in swallowing whole tablets often find it easier to swallow when the tablets are crushed and mixed into food vehicles such as applesauce.
- However, due to the hormone-based mechanism of action, crushing of the tablets should be avoided as it may cause unintended drug exposure to the caretakers.
- With its unique formulation, apalutamide disperses easily when mixed with applesauce, avoiding the need for crushing the tablets.

OBJECTIVES

- To determine the relative bioavailability of apalutamide administered as a mixture with applesauce vs whole tablets in healthy fasted male volunteers
- To confirm dose accuracy and in-use stability of apalutamide tablets in different food vehicles (applesauce, yogurt, orange juice, and green tea)

METHODS

Study design

- A total of 12 healthy men (aged 18-55 years) participated in this randomized, open-label, crossover phase 1 study conducted at a single center (NCT03802682).
- Participants were randomized (1:1) to receive apalutamide 240 mg as whole tablets (n=6) or as a dispersed mixture in applesauce (n=6) in fasted condition in a crossover manner separated by a washout period of 42-56 days between the two periods (or the reverse sequence).

Study drug administration

- Apalutamide tablets (4 × 60 mg) were dispersed in a specified amount of vehicle (i.e., 50 g or 200 g of applesauce or yogurt and 50 mL or 200 mL of orange juice or green tea extract).
- Uniform mixture was formed by gently mixing the contents with a spatula for about 10 seconds, repeated after 15 minutes and subsequently after an additional 15 minutes.
- Apalutamide-applesauce mixture was prepared by only trained pharmacist or pharmaceutical technician to ensure consistency.

METHODS

Assessments

- Participants were admitted to clinical facility on day -1 and remained there until after the collection of the 72-hour pharmacokinetic (PK) sample on day 4.
- Blood samples (2 mL each) for assessments of plasma concentrations of apalutamide were collected at predose and at 0.5, 1, 1.5, 2, 3, 4, 5, 6, 8, 12, 24, 36, 48, 60, 72 (day 4), 96 (day 5), 120 (day 6), 144 (day 7) and 168 hours (day 8) postdose.
- Plasma samples were analyzed to determine concentrations of apalutamide using a validated, specific, and sensitive liquid chromatography/ mass spectrometry/ mass spectrometry (LC-MS/MS) method.
- *In vitro* compatibility studies evaluated dose accuracy and in-use stability of apalutamide tablets when mixed into applesauce, yogurt, orange juice, and green tea.
 - ◊ For each vehicle, dose accuracy was assessed through Quality Control chromatography assay method performed in triplicate.
 - ◊ In-use stability testing included appearance of the mixture, assay of apalutamide, and chromatographic purity.

RESULTS

- Total exposure of apalutamide was comparable for apalutamide-applesauce mixture vs whole tablet administration.
- Demographics and baseline characteristics were generally comparable between two treatment sequence arms.

Table 1. Demographics and Baseline Characteristics

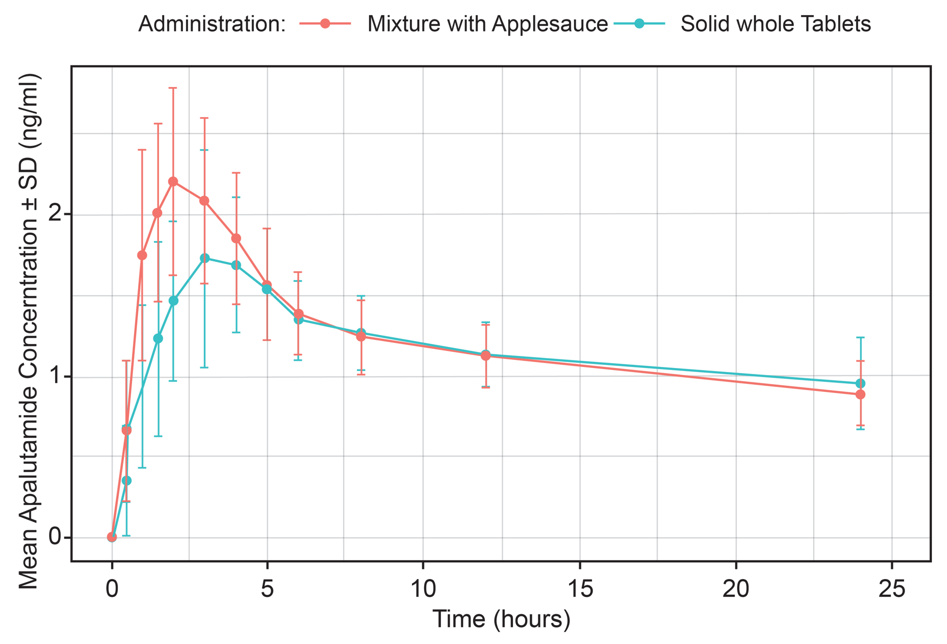
	Whole Tablets → Applesauce Mixture (n=6)	Applesauce Mixture → Whole Tablets (n=6)	Total (n=12)
Age, years	49.3 (3.8)	41.2 (13.5)	45.3 (10.4)
Male	6 (100)	6 (100)	12 (100)
White	6 (100)	6 (100)	12 (100)
Not Hispanic or Latino	6 (100)	6 (100)	12 (100)
Weight, kg	75.6 (10.9)	76.1 (11.7)	75.8 (10.8)
Height, cm	173.3 (7.1)	173.9 (7.5)	173.6 (7.0)
BMI, kg/m^2	25.1 (2.5)	25.1 (2.5)	25.1 (2.4)

Values are presented as mean (SD) or n (%) unless otherwise specified BMI, Body mass index

RESULTS

- Apalutamide plasma concentrations reached a maximum at 3 hours after administration of whole tablets and at 2 hours after administration of the applesauce mixture.
- On an average, apalutamide plasma levels were higher with the applesauce mixture compared to the whole tablets until about 4 hours postdose. Thereafter, the plasma concentrations were comparable in both treatments.
- Apalutamide plasma concentrations decreased in a multi-phasic manner and were quantifiable at the last PK sample (168 hours postdose) for both treatments and for all participants.

Figure 1. Mean Plasma Concentration-Time Profile of Apalutamide


Administration: Mixture with Applesauce Solid whole Tablets
Mean Apalutamide Concentration ± SD (ng/ml)
0
1
2
0
5
10
15
20
25
Time (hours)

- Total exposure as assessed by $AUC_{0\text{-}168h}$ increased by 5% after administration of apalutamide-applesauce mixture, with the 90% CI within the 80%-125% criterion for bioequivalence.
- Peak concentration (C_{max}) was increased by 27.6% for apalutamide-applesauce mixture compared to whole tablet administration with shorter median time (tmax) to reach C_{max}.

Table 2. Statistical Analysis of the Pharmacokinetic Parameters of Apalutamide

Parameter	n	Whole Tablets (Reference) Geometric Means	Applesauce Mixture (Test) Geometric Means	Applesauce Mixture vs Whole Tablets: Geometric Mean Ratio (%)	Applesauce Mixture vs Whole Tablets: Lower Limit; Upper Limit 90% CI (%)	Intra-subject CV (%)
C_{max} (µg/mL)	10	1.80	2.30	127.57	113.76; 143.05	13.8
$AUC_{0\text{-}72h}$ (µg.h/mL)	10	58.6	61.5	105.02	101.87; 108.27	3.7
$AUC_{0\text{-}168h}$ (µg.h/mL)	10	106	111	105.22	102.88; 107.60	2.7

$AUC_{0\text{-}72h}$, area under the plasma analyte concentration vs time curve from time 0 to 72 hours; $AUC_{0\text{-}168h}$, area under the plasma analyte concentration vs time curve from time 0 to 168 hours; C_{max}, maximum observed plasma analyte concentration; CV, coefficient of variance

- The compatibility studies found that apalutamide tablets were comparable in purity and dose accuracy when mixed in 50-200 g of applesauce or yogurt or in 50-200 mL of orange juice or green tea.
- Dose accuracy for all food vehicles was found to be within the criteria (limits: 85.0%-115.0%) and no food vehicles appeared to impact the availability of apalutamide.

Table 3. Comparison of Apalutamide Dose Accuracy in Different Food Vehicles

	Apalutamide (w/w%)[a]			
	Applesauce (n=12)	Yogurt (n=11)	Orange Juice (n=6)	Green Tea (n=11)
Control	102.3	102.6	102.0	102.0
Minimum	98.3	95.4	97.5	99.7
Maximum	104.5	101.3	102.0	102.3
Average	100.9	98.5	100.1	100.9

[a]Relative assay to reference assay analysis

Dose accuracy limits: No individual result outside 85.0%-115.0%

CONCLUSIONS

- **Apalutamide bioavailability ($AUC_{0\text{-}168h}$) was comparable between whole tablets and apalutamide-applesauce mixture; these results support apalutamide tablets (240 mg) dispersed in applesauce as a suitable alternative administration method.**
- **Alternative food vehicles of yogurt, orange juice, and green tea yielded similar results to applesauce in compatibility assessments of dose accuracy and purity of mixture.**

REFERENCES

1. Clegg NJ, et al. Cancer Res. 2012;72(6):1494-1503
2. ERLEADA® (apalutamide) tablets, for oral use. Full prescribing information; Available from: http://www.janssenlabels.com/package-insert/productmonograph/prescribing-information/ERLEADA-pi.pdf

DISCLOSURES

Alex Yu, Maura Erba, Rushikesh Potdar and Anasuya Hazra are employees of Janssen Research and Development (a subsidiary of Johnson & Johnson) and own Johnson & Johnson stock or stock options.

This study was funded by Janssen Research & Development, LLC
Previously presented at AAPS PharmSci 360, October 26-November 05, 2020

Poster Presented at National Association of Specialty Pharmacy, Washington, DC, September 27-30, 2021